Answer Key — Competitive Exam Practice 9: Priority, Round Robin, Multilevel Queues, and Multiprocessor Scheduling

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Concepts That Click

Q1 (B). No process ever exhausts its quantum, so none is preempted and the circular queue is served in arrival order — exactly FCFS.

Q2 (B). (A) describes MLFQ, not MLQ, whose queue assignment is permanent. (C) is false: a steady stream of higher-priority arrivals can starve a low-priority process. (D) is false: average WT is not monotone in q (on the lecture workload, $q = 2, 3, 4$ give 13.0, 14.6, 13.0 ms).

Q3 4.00. Gantt: P1 0–1, P2 1–4 (preempts P1), P1 4–8, P4 8–10, P3 10–18. Completions 8, 4, 18, 10; TAT $= 8, 3, 16, 7$; WT $= 3, 0, 8, 5$; average $16/4 = 4.00$.

Q4 8.25. Gantt: P1 0–2, P2 2–4, P3 4–5, P1 5–7, P4 7–9, P2 9–10, P1 10–11, P4 11–13. At $t = 2$, P2 and P3 have arrived and join before the preempted P1: [P2, P3, P1]; at $t = 4$, P4 (arrived at 3) joins before the preempted P2: [P3, P1, P4, P2]. Completions 11, 10, 5, 13; TAT $= 11, 9, 3, 10$; average $33/4 = 8.25$.

Q5 12. P1 0–2 (Q_0, demoted), P2 2–4 (Q_0, demoted), P1 4–5 (Q_1) preempted by P3's arrival, P3 5–6 (Q_0, done), P2 6–8 (Q_1, done), P1 8–12 (its remaining 4 ms). The answer is 12 whether or not a preempted process keeps the rest of its quantum.

Q6 180. $(n - 1)q = 9 \times 20 = 180$ ms.

Q7 80. $q/(q + s) = 4/5 = 80\%$.

Q8 14. CPU0: J1 0–8, J4 8–10, J6 10–14. CPU1: J2 0–3, J3 3–9, J5 9–14. Both finish at 14 $= 28/2$, the lower bound.

Q9 (C). In asymmetric multiprocessing a single master CPU makes all scheduling decisions; self-scheduling on shared kernel data describes SMP. (A), (B), (D) are true.